

STAGE 2 SUBMISSION - 1-PAGE EXECUTIVE SUMMARY

Applicant: Dipta Dhor | Track: Track E - AI/ML: Auto-Marking

1. Project Overview & Deliverables Completed

To meet CS Revise's requirement for automated assessment across diverse GCSE and 11+ curriculum formats, I engineered a production-ready multimodal marking engine (`auto_marker.py`) powered by Gemini 3.6 Flash via Google AI Studio's free tier, enforcing strict Pydantic-backed JSON output schemas. The system comprehensively handles:

- **Handwritten Short Answers:** Direct vision ingestion of photographed student handwriting (e.g. human circulatory system) evaluating scientific understanding despite minor phonetic misspellings.
- **Handwritten Diagram Logic:** Full visual structure evaluation of hand-drawn student flowcharts (e.g. even/odd logic), validating shapes, decision diamonds, branches, and control flow.
- **Extended Long-Form Essays:** Multi-tier qualitative breakdown (AO1 Knowledge, AO2 Application, AO3 Analysis/Evaluation) applied to complex scenarios like school cloud database migration.
- **Standard Formats & Reporting:** Deterministic handling of MCQs, automated multi-case demo runs (`run_demo.py`), and human-readable audit export (`grading_report.txt`).

2. Engineering Approach & Safeguards

After benchmarking keyword/n-gram matching (too brittle for synonyms) and cosine embedding similarity (unable to enforce discrete partial-mark rubrics), an end-to-end Multimodal Vision-Language architecture was chosen:

- **Direct Multimodal Vision Ingestion:** Eliminates decoupled OCR errors by feeding raw submission photos and sketches directly into Gemini's multimodal attention space, natively reading logic and handwriting.
- **Pydantic Schema Enforcement:** Structured JSON contracts guarantee typed responses (breakdown, itemized justifications, strengths, improvement areas) with zero reliance on fragile prompt regex.
- **Local Arithmetic Reconciliation:** Post-inference validation programmatically verifies `sum(marks_awarded) == total_score`, auto-clamps out-of-range marks, and flags mathematical anomalies for human review.
- **Resilience & Fallbacks:** Exponential backoff accommodates free-tier rate limits (429/503). Critical failures fail-safe to zero marks with human-review escalation rather than returning hallucinated grades.

3. Key Assumptions & Operational Strategy

- **API Environment & Cost Optimization:** Assumes zero API expenditure for the prototype by operating on Google AI Studio's free tier. For live deployment, request throttling or provisioned enterprise endpoints will handle peak exam surges.
- **Legibility & Explicit Uncertainty:** Assumes the engine must transcribe handwriting verbatim or flag visual ambiguity explicitly in notes rather than guessing unreadable student work.
- **Platform Integration Architecture:** Assumes a stateless callable contract (`GradingRequest` -> `GradeOutput`), allowing seamless integration into AWS Lambda, Google Cloud Functions, or async task queues.